

Concerns about surveillance overreach and misuse of geospatial data for individual tracking require policy safeguards and end-to-end encryption. Measures must prevent the injection of objectionable material into broadcasts. Additionally, stringent cybersecurity regulations, failsafe redundancy, and regular penetration testing are essential to mitigate data breaches and cybercrimes.

The global impact of the D2M technology

While India leads in D2M development, other countries are adopting it for media delivery and crisis

communication. The UK, Finland, Germany, and the Czech Republic have conducted trials for terrestrial broadcasting services. China recently demonstrated real-time transmission of 360-degree panoramic 4K HD video via D2M. Similarly, South Korea showcased live UHD content delivery, and Italy trialled D2M during the Eurovision Song Contest 2022, highlighting its potential for audience engagement. A high-quality broadcast signal was transmitted simultaneously live from sites in four European cities. Users with D2M broadcast-enabled smartphones were able to watch the

simultaneously, reflecting growing global interest.

The way ahead

D2M technology offers a transformative approach to streaming by converging terrestrial transmitters, satellite broadcasting networks, and broadband services. It offloads high-demand video content from cellular to broadcast networks, easing capacity constraints. The technology promises enhanced customer reach for broadcasters and facilitates real-time dissemination of verified alerts during emergencies. It also enables the creation of sleeker, cost-effective devices using fewer resources, benefiting sectors like education, finance, health, and livelihoods, even in remote regions.

However, the technology's success depends on addressing challenges such as infrastructure investments, equipment interoperability, device compatibility, spectrum availability, and regulatory frameworks. The government and regulators are yet to finalise the policy framework and regulations for: providing a level playing field, public funding, network integration, quality control, and intellectual property rights protection. It demands intensive research across technology and business models. Technological areas, such as transmission protocols, modulation schemes, and video encoding and multiplexing require an in-depth study.

The technology demands a different skillset, as the professionals need to understand mobile communication protocols as well as the content production. Modalities need to evolve for awareness and adaptation of the technology across urban and rural demographics. While D2M technology presents a number of exciting possibilities, close coordination among stakeholders is essential to deliver an affordable, attractive solution for end-users, broadcasters, and mobile device manufacturers alike. **EFY**

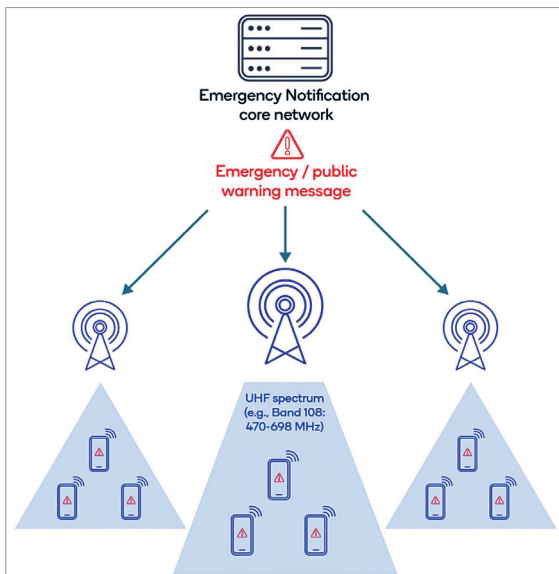


Fig. 4: Delivering emergency notifications during disaster events (Source: IIT Kanpur, June 2022 white paper)

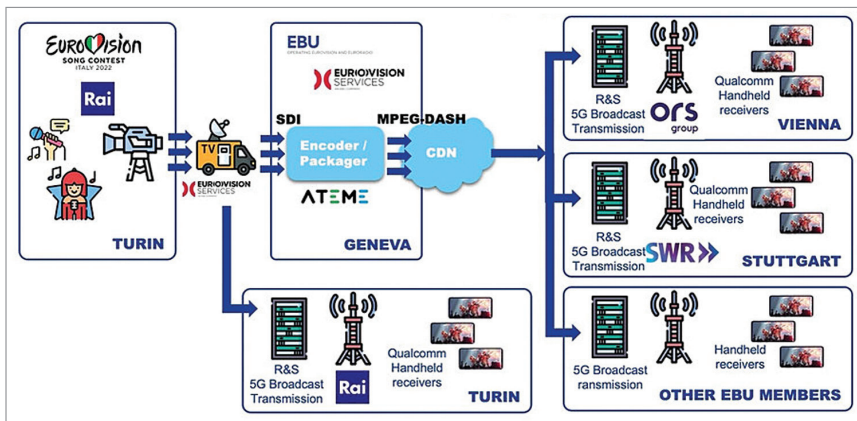


Fig. 5: Eurovision song contest 5G broadcast trials (Source: <https://www.5g-mag.com>)